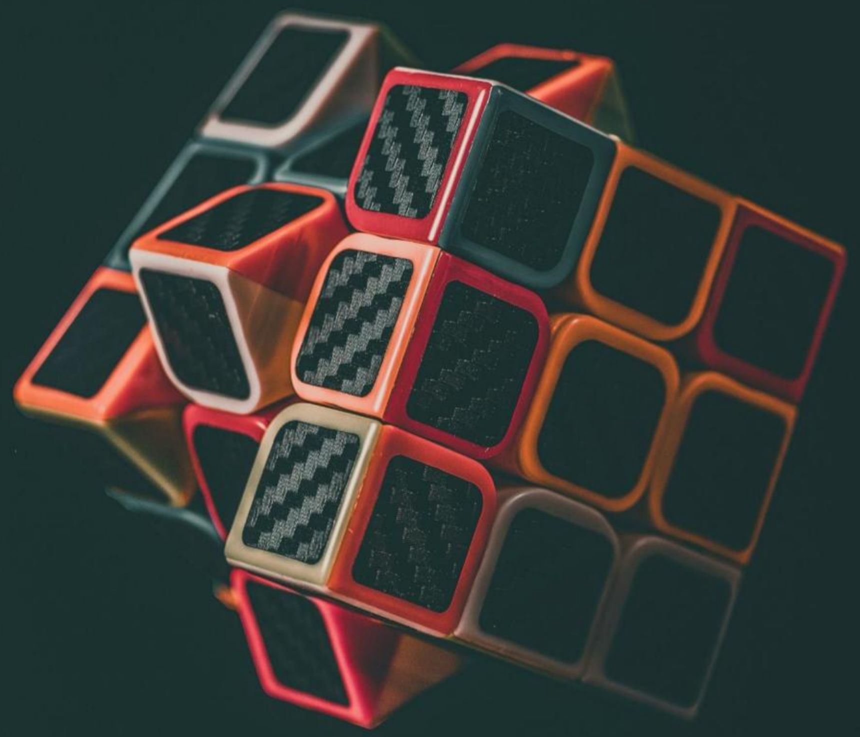
UTILITY FUNCTIONS FOR OPTIMIZATION



Mathematical Concepts of Machine Learning Seminar

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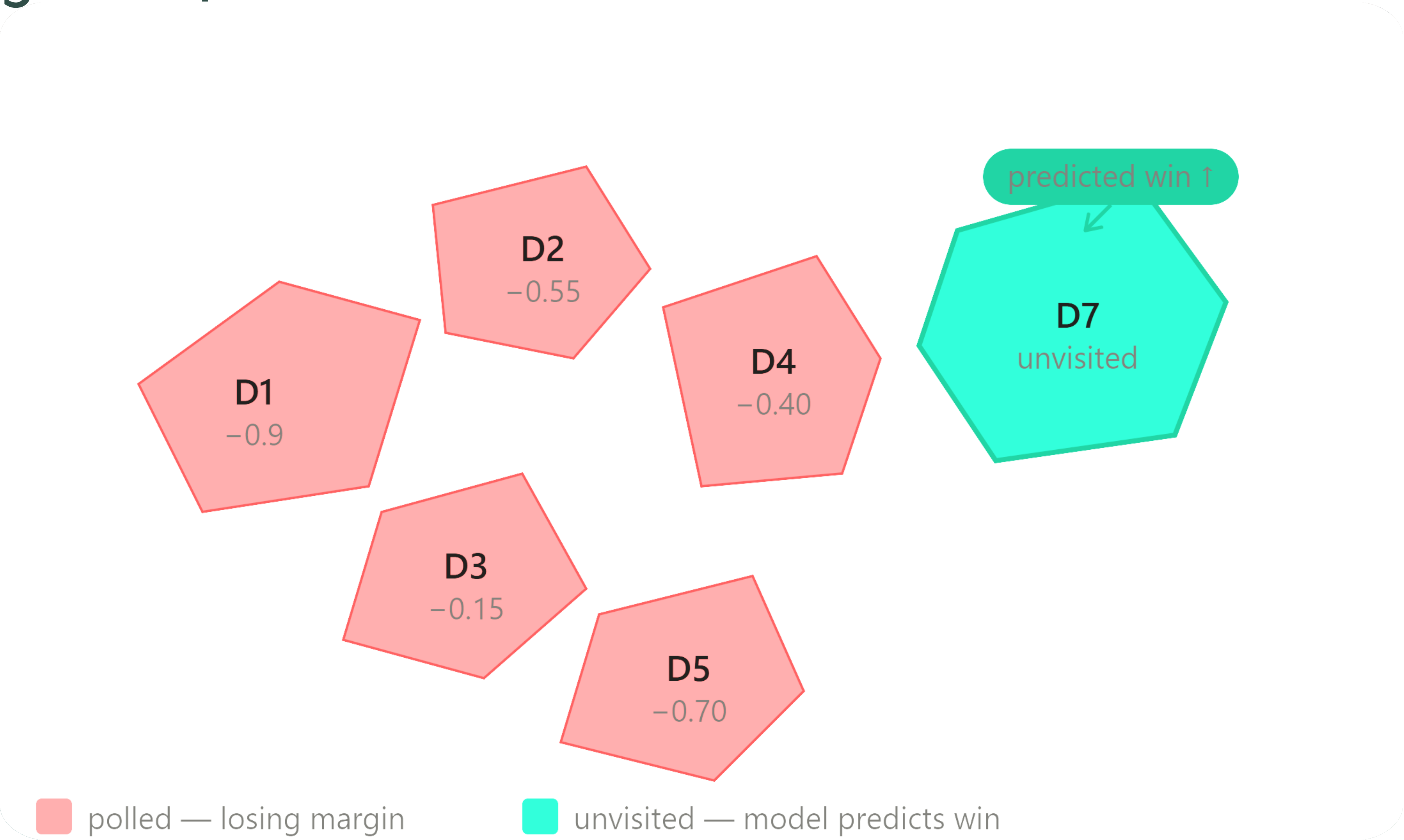
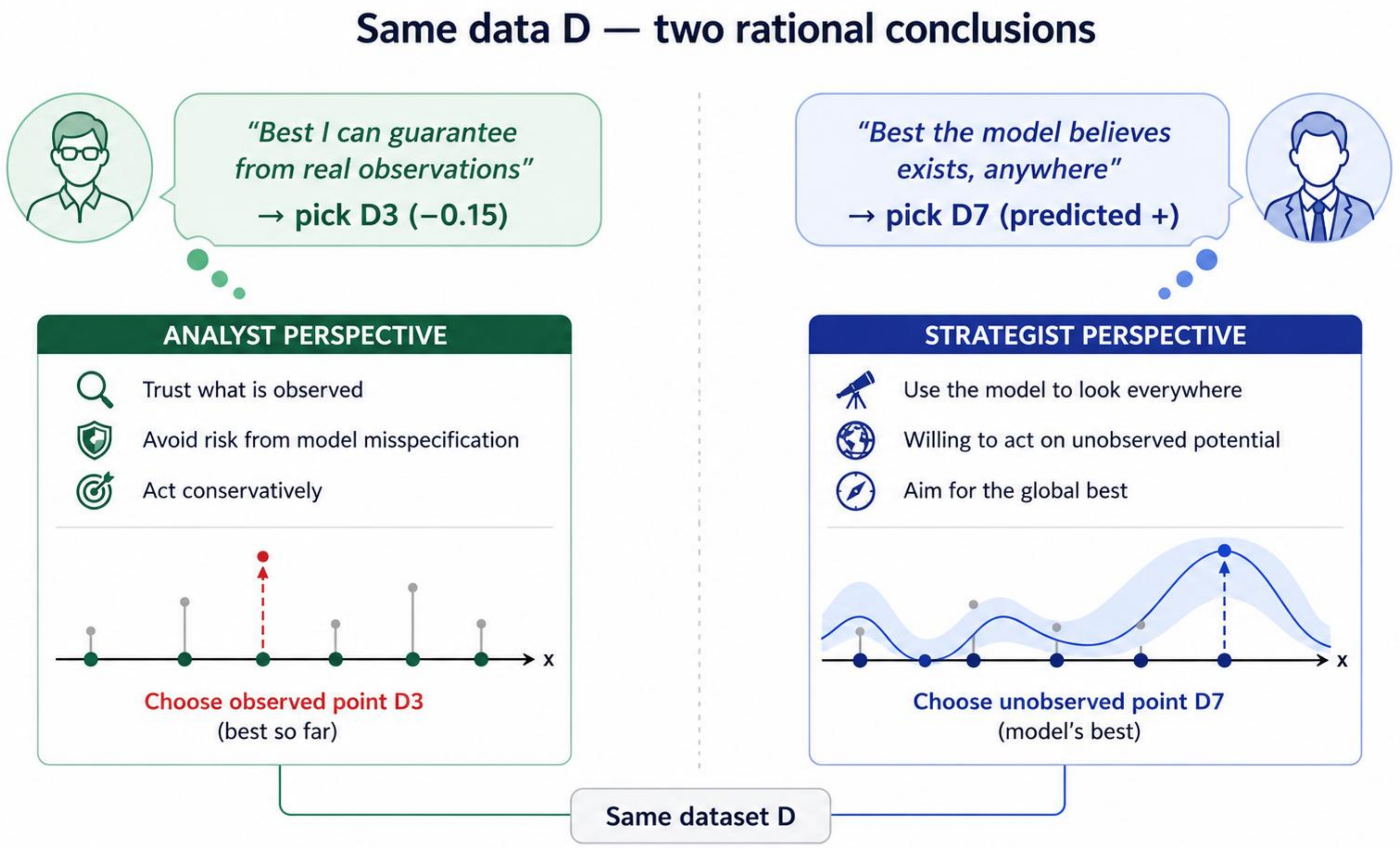
Opening Example




Same data D — two rational conclusions
"Best I can guarantee from real observations"
→ pick D3 (−0.15)
"Best the model believes exists, anywhere"
→ pick D7 (predicted +)
ANALYST PERSPECTIVE
Trust what is observed
Avoid risk from model misspecification
Act conservatively
x
Choose observed point D3
(best so far)
STRATEGIST PERSPECTIVE
Use the model to look everywhere
Willing to act on unobserved potential
Aim for the global best
x
Choose unobserved point D7
(model's best)
Same dataset D

Motivation

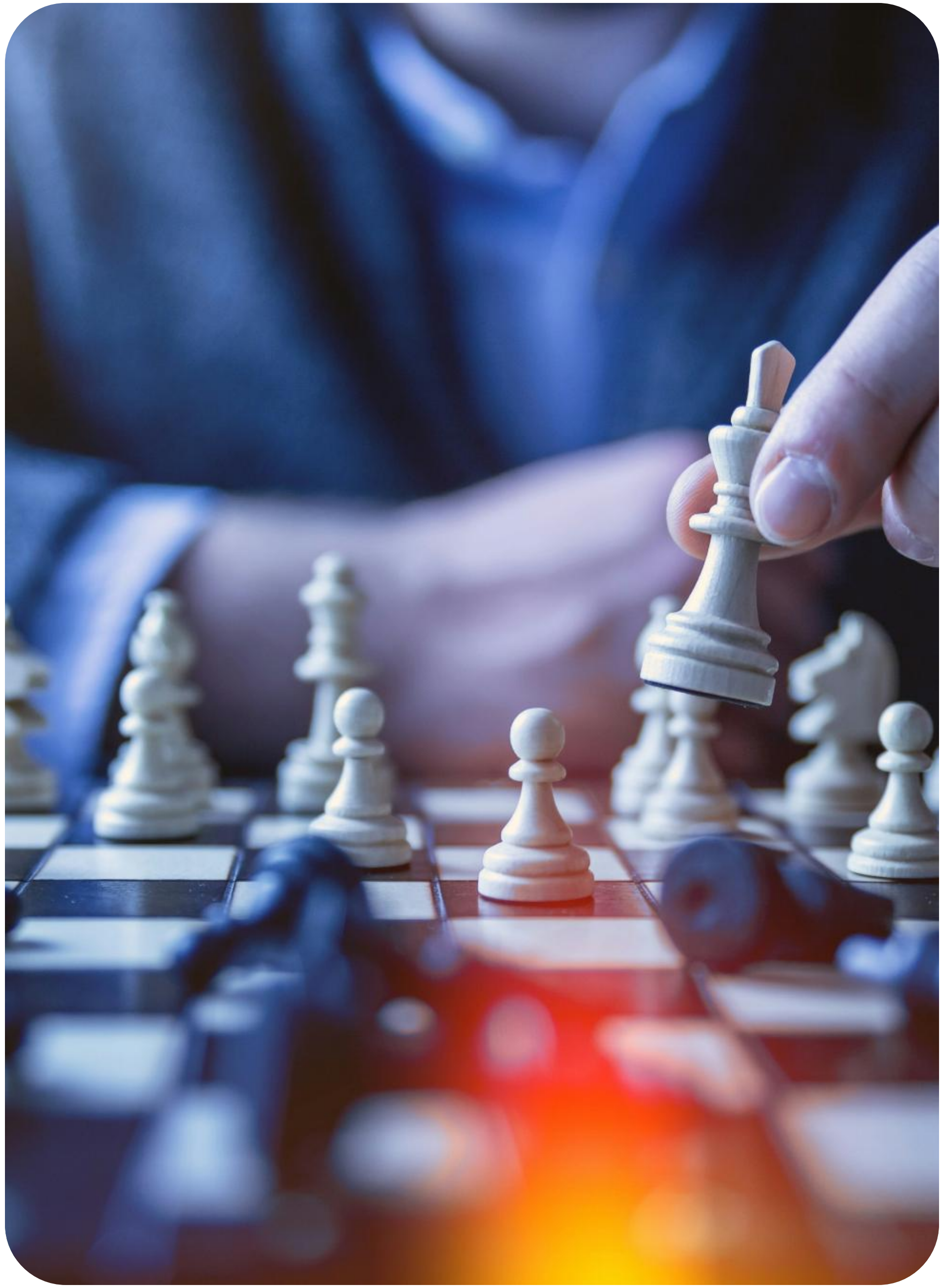
- Bayesian optimization needs to decide *what to observe next*
- A model alone doesn't tell you what to do, you need to define what "success" means
- The utility function u(D) does exactly that: it scores how good a collected dataset is
- Everything in BO flows from this choice

The setup

- We run optimization, then commit to one final point x* permanently
- Example: final drug dosage, last campaign district, chosen hyperparameter
- All observations only matter insofar as they help us choose x* well
- We pick x* to maximize expected utility:

$$x^* \in \operatorname*{argmax}_{x' \in A} \; \mathbb{E}[\, \upsilon(\varphi') \mid x', D \,]$$

expected utility of recommending x′, given data D



The Core Formula

- Collapsing the decision into a dataset-level utility:

$$u(D) = \max_{x' \in A} \mathbb{E}[\, \upsilon(\varphi') \mid x', D \,]$$

best expected utility over all candidate recommendations x′

- This is the central equation of the chapter
- Two design choices remain: action space A, and terminal utility $\upsilon(\phi)$

Action Space

- Where are we allowed to recommend after optimization?
- Visited points only → safe, grounded in real observations
- Entire domain X → can recommend unvisited points, but trusts the model
- This one choice leads to two completely different utility functions

Risk Tolerance

- The shape of $u(\phi)$ encodes your attitude toward uncertainty
- Linear → risk neutral (standard in BO)
- Concave → risk averse: certainty equivalent $<$ expected value (Jensen's inequality)
- Convex → risk seeking: certainty equivalent $>$ expected value
- Practical form: $\mu + \beta\sigma$ where $\beta < 0$ averse, $\beta = 0$ neutral, $\beta > 0$ seeking

Simple Reward

- Risk neutral + action space restricted to visited points
- $u(D) = \max \mu\_D(x)$ best posterior mean at observed locations
- With exact observations: $u(D) = \max \phi$ just the best value seen
- One-step lookahead → Expected Improvement (EI)
- Local, safe, model-agnostic

Global Reward

- Risk neutral + action space = entire domain X
- $u(D) = \max_{x \in X} \mu_D(x)$ global max of the posterior mean
- Can recommend unvisited points, trusts the model
- One-step lookahead → Knowledge Gradient
- Global, riskier, more powerful if model is good

Simple vs Global: The Key Distinction

- Simple reward = local. Only looks at where you've been.
- Global reward = global. Looks at what the model believes everywhere.
- They can strongly disagree, and that disagreement is meaningful

Visualization

Nonsensical Alternative

- Sometimes seen in literature: $u(D) = \max y$
- Looks like simple reward — but uses raw noisy observations, not posterior mean
- The problem: rewards datasets with lucky outliers, not genuinely good points
- A noisy spike on a bad day looks better than a consistent win
- Posterior mean μ_D correctly discounts noise — max y does not
- Works as approximation only when noise is very low

Cumulative Reward

- What if every observation matters, not just the best one?
- $u(D) = \Sigma y_i$ — sum of all observed values
- Right when the optimization process itself has consequences
- Example: active search — find as many winning districts as possible, everyone counts
- Foundation for regret analysis and bandit algorithms (Chapter 7)

Information Gain

- Entirely different philosophy: measure what you learned, not what you found
- Choose a quantity of interest ω (e.g. location x^* or value f^* of the optimum)
- $u(D) = H[\omega] - H[\omega \mid D]$ — reduction in entropy after seeing data
- Equivalently: $u(D) = D\_KL(p(\omega \mid D) \parallel p(\omega))$
- Both definitions give the same policy (equal expected values — proof p.138)
- One-step lookahead → Entropy Search family of algorithms

Comparison

Utility	Action space	→ Acquisition function
Simple reward	Visited points	Expected Improvement (EI)
Global reward	Entire domain	Knowledge Gradient
Cumulative reward	–	GP-Bandits / UCB
Info gain (x*)	–	Entropy Search
Info gain (f*)	–	Predictive Entropy Search

Key Takeaways

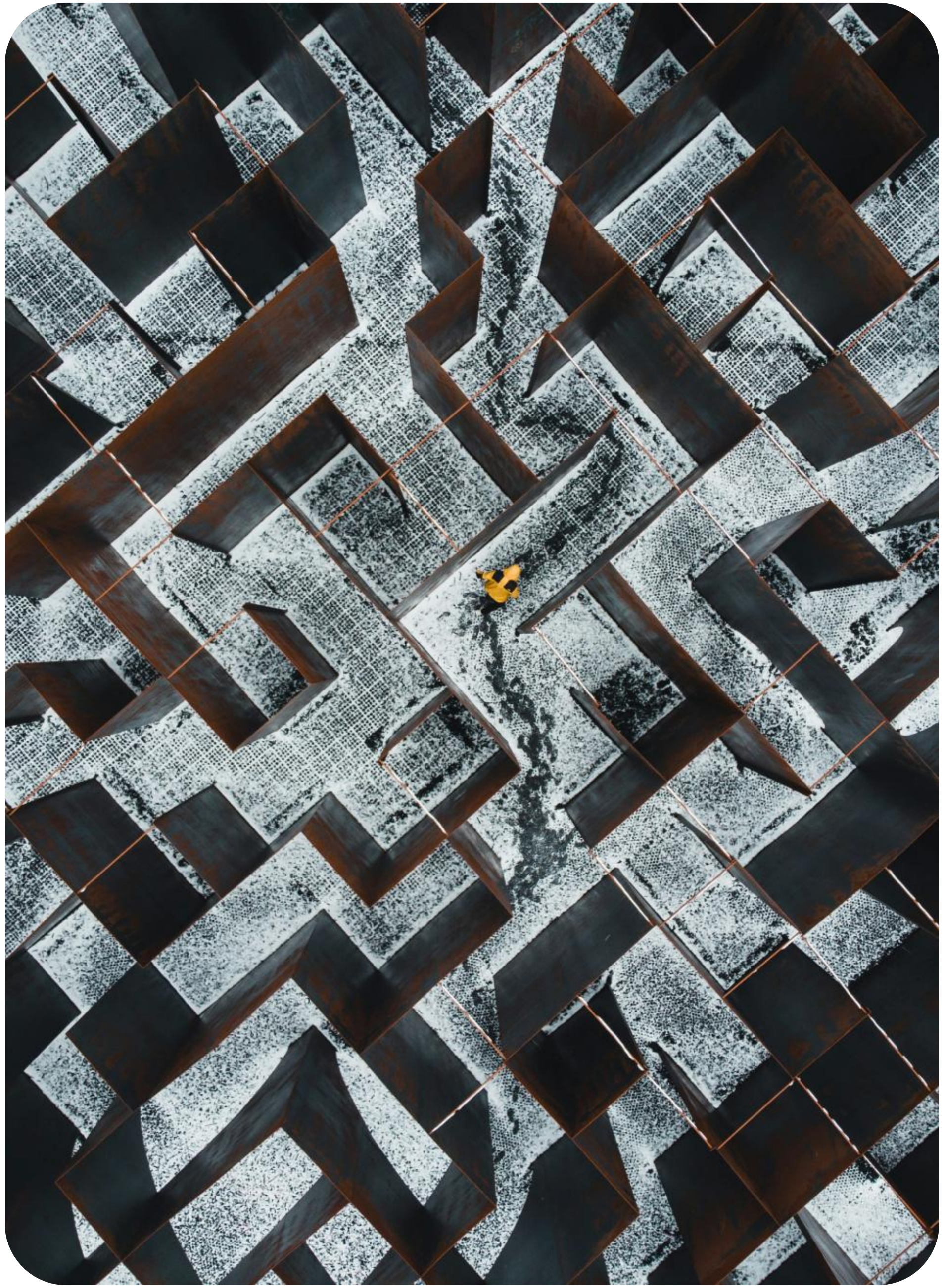
Utility functions are not cosmetic — they determine the entire policy

Simple vs global reward = local safety vs model-trusting ambition

Risk tolerance is a real design choice, not a default

Information gain reframes BO as *learning*, not just optimizing

The "right" utility depends on your problem — knowing the tradeoffs is the skill



THANKS FOR YOUR ATTENTION!